

Context from the preceding pages.

The experiments evaluate $\text{sys}(K) = c_{\text{EHZ}}(K)^2/(2 \text{vol}_4(K))$ numerically. For a four-dimensional polytope, R is the sum of its ridges' absolute symplectic areas divided by $\sqrt{\text{vol}_4(K)}$, and C is the largest ridge area divided by their sum. The preceding discussion found an inverse association between R and the systolic ratio. A $k \times m$ product has polygonal factors with k and m sides. The preceding comparison gives $R = 8$ and $\text{sys} = 1/2$ for the cube, versus $R \approx 8.944$ and $\text{sys} > 1$ for HKO.

1 Geometric information about large systolic ratios

Within the sampled product distributions, low normalized ridge area and balanced symplectic vertex covariance each identify populations with larger mean numerical systolic ratios. These give two different geometric descriptions of promising bodies: one through their two-dimensional faces, the other through their extreme points. Both respect affine symplectic changes of coordinates and can be evaluated before capacity. The evidence supports geometric selection; the square–HKO comparison above shows why it cannot turn small ridge area into a universal rule for increasing the ratio of an individual body.

The covariance condition makes the second description precise. For vertices $v_1, \dots, v_N$, put

$$\bar{v} = \frac{1}{N} \sum_{i=1}^N v_i, \quad \Gamma = \frac{1}{N} \sum_{i=1}^N (v_i - \bar{v})(v_i - \bar{v})^T.$$

In suitable symplectic coordinates, this matrix has Williamson normal form $\text{diag}(\nu_1, \nu_2, \nu_1, \nu_2)$, with $0 < \nu_1 \leq \nu_2$. The ratio $\rho(K) = \nu_2/\nu_1$ measures the imbalance between the covariance scales of the two symplectic coordinate planes. It equals one when those scales agree.

This score is invariant under affine symplectic maps and common dilation. Indeed, centering removes translations, and a symplectic map S gives $\Gamma(SK) = S\Gamma(K)S^T$. With $J = \begin{pmatrix} 0 & I_2 \\ -I_2 & 0 \end{pmatrix}$,

$$J\Gamma(SK) = S^{-T}(J\Gamma(K))S^T.$$

Thus the eigenvalues $\pm i\nu_1, \pm i\nu_2$ of $J\Gamma$ are unchanged. A dilation multiplies both ν_j by the same squared scale factor, preserving their ratio.

For a product, if A and B are the planar vertex covariance matrices of its factors, then $\Gamma = \text{diag}(A, B)$. Consequently $(J\Gamma)^2 = \text{diag}(-BA, -AB)$, so ν_1^2, ν_2^2 are the eigenvalues of AB .

The following contrasts give the empirical support for these conditions. Each compares selections within the same generated population and factor-side-count groups; the bodies were selected from their geometry before their ratios were evaluated.

Selection	Compared with	Mean difference
Low R	Comparison bodies	+0.311
Lower C at low R	Higher C at low R	+0.034
Low ρ	Comparison bodies	+0.333
Low ρ	Low R , then lower C	+0.015

The first contrast has selected and comparison means 0.626 and 0.316. The second splits the lowest one percent by R into halves according to C ; its difference is positive in eight of ten side-count groups. Thus the dominance of the largest ridge adds information even when total ridge area is already small.

The two covariance contrasts come from a separate comparison on two fresh pools, motivated by an exploratory inverse association between ρ and the ratio. Their descriptive 95% Student- t intervals, using the twenty pool-by-group differences, are $[0.310, 0.356]$ and $[-0.015, 0.044]$. These contrasts support both selection criteria, without resolving which yields larger mean ratios. The geometric reason for their association with capacity remains open.

1.1 Changing the population and strengthening the filter

Ridge area also helped select generic ten-facet polytopes. Here the number of ridges varies even though the number of facets is fixed. To compare the area per ridge, we used R divided by the ridge count. The lowest one percent by this quantity had mean systolic ratio 0.626, compared with 0.339 for comparison bodies from the same pool. More severe filtering gave no further improvement in this sample: the lowest tenth of the selected group had a mean ratio 0.032 below the remaining selected bodies. The descriptive 95% interval from resampling these selected bodies was $[-0.095, 0.033]$, which spans zero. This interval conditions on the realized selection; it does not include variation from generating and selecting a new pool.

The product filters also remained useful when the factors were tangential polygons. These polygons have every side tangent to a circle. We obtained them by keeping the sampled unit normals and setting every support height to one, retaining pairs for which both the original and the changed polygons were valid. For products with side counts 4×6 and 6×6 , selection by small ρ increased the mean ratio by 0.259 relative to comparison bodies; ridge selection increased it by 0.225. These are averages of the two group differences in one pool. Together with the original product experiment, they show that the usefulness of both filters survived this particular change of sampling distribution.

2 Changing a body or changing the sampling distribution

Selection uses geometry to choose among sampled bodies. We can also ask whether changing that geometry raises the ratio of an individual body, or whether the successful samples suggest a better sampling distribution.

2.1 Making the factors tangential

Replacing both factors by tangential polygons sometimes raised and sometimes lowered the systolic ratio. With the normals fixed, setting all support heights to one gave a mean numerical increase of 0.015 among the tested 4×4 and 4×6 products, with changes of both signs. Thus a useful selection rule for tangential products does not supply a rule for improving every product by making its factors tangential.

This comparison changes the support heights, then scales each factor to area one. The latter normalization preserves the systolic ratio: for $a, b > 0$,

$$\text{diag}(aI_2, bI_2) = \sqrt{ab} \text{diag}(\sqrt{a/b} I_2, \sqrt{b/a} I_2),$$

a common dilation followed by a symplectic map. The comparison concerns products for which the original factors and both height replacements remain admissible; it measures the change of shape within that family.

The tested ambient rotations $K \mapsto UK$, $U \in SO(4)$, also gave smaller search maxima than fresh sampling at equal evaluation budgets (Appendix A).

2.2 Learning a sampling distribution

Adapting the sampling distribution improved the search for pentagon products in each of three comparisons. The cross-entropy method [Boe+05] repeatedly fits a distribution to the best part of the current population and draws a new population from it. Under the same number of capacity evaluations, it attained both a larger maximum systolic ratio and a larger median among the eight best distinct bodies:

Run	Maximum		Median of best eight	
	Independent	Adaptive	Independent	Adaptive
1	0.760	0.799	0.729	0.782
2	0.811	0.831	0.754	0.776
3	0.760	0.856	0.722	0.814

The gain therefore extended beyond a single best observation in each run.

The sampler described each factor by log ratios of its normal-angle gaps and centered logarithmic lengths of its dual normals, together with the relative angle between the factors. These coordinates remove separate factor scales and simultaneous planar rotation, which preserve the systolic ratio. Independent Gaussian coordinates described gaps and lengths; the relative angle used a wrapped Gaussian with a circular mean update. Each update shifted the distribution’s moments towards those of the quarter of the population with largest ratios.

The comparison also includes the numerical evaluator’s input restrictions. It accepts only bodies within a prescribed coordinate range. Independent sampling uses the original coordinates, whereas the adaptive method reconstructs a normalized representation, so this restriction affects the two distributions differently. The observed gain belongs to the implemented combination of adaptation and admission. All recorded ratios remained below one.

A Ambient rotations and fresh sampling

The ambient action

$$P \times Q \mapsto U(P \times Q), \quad U \in SO(4),$$

preserves Euclidean shape and volume. Capacity can change because U need not preserve the symplectic form; the image need no longer be a Lagrangian product for the fixed form.

Fresh sampling attained a larger maximal numerical ratio than the tested rotations in each of four comparisons with equal numbers of capacity evaluations. The searches shared their starting body. Rotation improved three of the four starts. For example, a quadrilateral product improved from 0.328 to 0.529 under the tested rotations, while fresh sampling reached 0.774. These results concern the prescribed rotation schedule and starting products; they do not compare all ambient rotations with all products from the sampling law. The schedule and complete allocation are retained with the experiment’s methods.

References

- [Boe+05] Pieter-Tjerk de Boer et al. “A Tutorial on the Cross-Entropy Method”. In: *Annals of Operations Research* 134 (2005), pp. 19–67. URL: <https://people.smp.uq.edu.au/DirkKroese/ps/aortut.pdf>.